

Here is a **complete GitHub Actions course outline (DevOps + Interview + Real-world ready)** — structured step-by-step so you can **learn → build → master**.

✓ GitHub Actions Complete Course Outline

◆ Module 1: Introduction to GitHub Actions

- What is CI/CD
 - What is GitHub Actions
 - Why GitHub Actions vs Jenkins/GitLab CI
 - Core components:
 - Workflow
 - Job
 - Step
 - Runner
 - Event
 - Basic workflow lifecycle
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◆ Module 2: Workflow YAML Fundamentals

- File location:
- `.github/workflows/*.yml`
- Basic syntax:

name:

on:

jobs:

 job-name:

 runs-on:

 steps:

- Common keys:
 - name
 - on

- jobs
 - steps
 - uses
 - run
 - env
 - defaults
-

◆ Module 3: Events (Triggers)

- push
 - pull_request
 - workflow_dispatch (manual)
 - schedule (cron jobs)
 - repository_dispatch
 - release
 - tag push
 - Filtering:
 - branches
 - paths
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◆ Module 4: Runners

- GitHub-hosted runners
 - Self-hosted runners
 - Runner OS:
 - ubuntu
 - windows
 - macOS
 - Customizing runners
 - Runner labels
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◆ Module 5: Jobs & Steps Deep Dive

- Multiple jobs
- Job dependencies:

needs:

- Parallel vs sequential jobs
 - Steps:
 - run commands
 - use actions
 - Job outputs
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◆ Module 6: Actions (Core Concept)

- What are actions
 - Types:
 - Marketplace actions
 - Custom actions
 - Using actions: uses: actions/checkout@v4
 - Inputs & outputs
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◆ Module 7: Expressions & Contexts (VERY IMPORTANT)

- Syntax: `${{ }}`
 - Contexts:
 - github
 - env
 - job
 - steps
 - runner
 - secrets
 - Conditionals: `if: github.ref == 'refs/heads/main'`
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◆ Module 8: Environment Variables & Secrets

- env usage
- Default variables:
 - GITHUB_SHA
 - GITHUB_REF
- Secrets:
 - Repository secrets
 - Org secrets
- Using secrets securely:

```
${{ secrets.MY_SECRET }}
```

◆ Module 9: Workflow Inputs & Outputs

- Inputs with workflow_dispatch
 - Passing data between steps: echo "name=value" >> \$GITHUB_OUTPUT
 - Job outputs
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◆ Module 10: Artifacts & Cache

- Upload artifacts: actions/upload-artifact
- Download artifacts
- Cache dependencies:

actions/cache

- Cache keys strategy
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◆ Module 11: Matrix Strategy (Powerful)

- Run jobs in parallel:

strategy:

matrix:

python: [3.8, 3.9, 3.10]

- fail-fast

- include/exclude
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◆ **Module 12: Reusable Workflows**

- Reuse workflows across repos: workflow_call:
 - Passing inputs & secrets
 - DRY principle in CI/CD
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◆ **Module 13: Custom Actions**

- Types:
 - JavaScript actions
 - Docker actions
 - Composite actions
 - action.yml structure
 - Inputs / outputs
 - Publishing actions
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◆ **Module 14: Security Best Practices (Interview MUST)**

- Secrets handling
 - Least privilege
 - Avoid hardcoding
 - OIDC authentication (AWS/Azure/GCP)
 - Masking secrets
 - Secure workflows
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◆ **Module 15: GitHub Actions + Docker**

- Build image
- Login to DockerHub
- Push image
- Multi-stage builds

- Example pipeline
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◆ Module 16: GitHub Actions + AWS (Production)

- AWS login:
 - Access keys
 - OIDC role (IMPORTANT)
 - Services:
 - ECR (push image)
 - ECS / EKS deployment
 - S3 upload
 - Terraform integration
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◆ Module 17: GitHub Actions + Kubernetes

- Deploy using kubectl
 - Helm deployments
 - Rolling updates
 - Rollbacks
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◆ Module 18: CI/CD Pipeline Design (End-to-End)

- Build → Test → Scan → Package → Deploy
 - Multi-stage pipelines
 - Branch strategy:
 - dev
 - staging
 - prod
 - Environment approvals
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◆ Module 19: Deployment Strategies

- Blue-Green deployment

- Canary deployment
 - Rolling deployment
 - Feature flags
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◆ Module 20: Debugging & Troubleshooting

- Debug logs: ACTIONS_STEP_DEBUG: true
 - Common errors:
 - permission denied
 - secret issues
 - runner issues
 - Re-run jobs
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◆ Module 21: Monitoring & Notifications

- Workflow status badges
 - Slack integration
 - Email alerts
 - Custom notifications
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◆ Module 22: Advanced Topics

- Concurrency: concurrency:
 - Timeout
 - Conditional jobs
 - Dynamic workflows
 - Workflow chaining
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◆ Module 23: Real-World CI/CD Projects (Must Practice)

1. ☒ Node.js app pipeline
2. ☒ Python app pipeline

3. ☒ Docker build & push
 4. ☒ Deploy to EKS using Helm
 5. ☒ Terraform infra deploy
 6. ☒ Multi-env pipeline
 7. ☒ Secure AWS OIDC pipeline
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◆ Module 24: Interview Preparation

- Difference:
 - GitHub Actions vs Jenkins
 - Workflow explained
 - OIDC vs Access keys
 - Matrix & cache explained
 - Debugging scenarios
 - Real CI/CD architecture design
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◆ Module 25: Best Practices Cheat Sheet

- Keep workflows modular
 - Use reusable workflows
 - Minimize secrets exposure
 - Use caching
 - Write readable YAML
 - Use versioned actions:
 - @v4 (NOT @main)
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☒ BONUS: Minimal Working Example

name: CI Pipeline

on:

push:

branches: [main]

jobs:

build:

runs-on: ubuntu-latest

steps:

- name: Checkout code

uses: actions/checkout@v4

- name: Run script

run: echo "Hello GitHub Actions"
